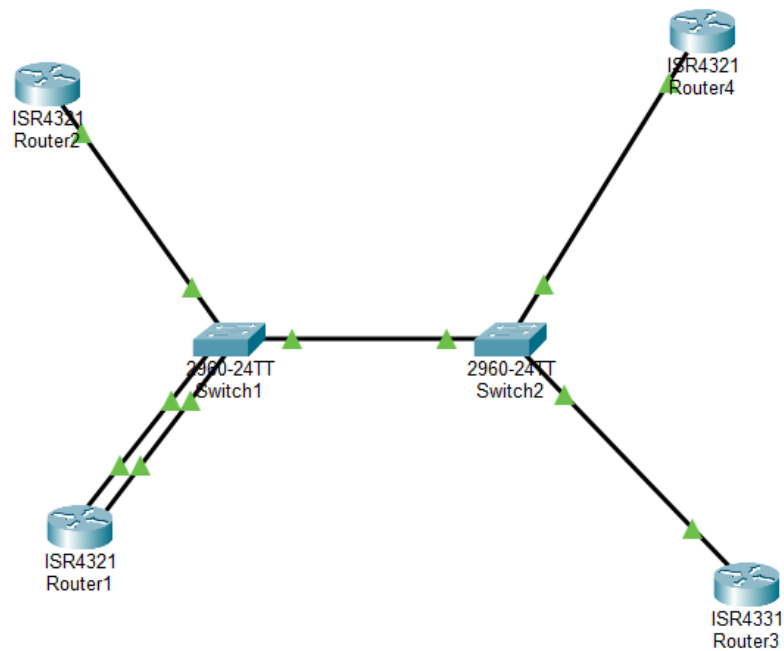


Week 2: Cisco Device Functions

Task 5



```
Switch>show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       0001.9629.0a18    DYNAMIC   Fa0/24
1       000a.f320.be01    DYNAMIC   Fa0/2
1       00e0.b000.e201    DYNAMIC   Fa0/1
1       00e0.b000.e202    DYNAMIC   Fa0/3
Switch>clear mac address-table
      ^
% Invalid input detected at '^' marker.
Switch>clear mac adress-table
      ^
% Invalid input detected at '^' marker.
Switch>show mac adress-table
      ^
% Invalid input detected at '^' marker.
Switch>
```

Task 6

```

Router#
Router#show interface
GigabitEthernet0/0/0 is up, line protocol is up (connected)
  Hardware is Lance, address is 00e0.b000.e201 (bia 00e0.b000.e201)
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Full-duplex, 100Mb/s, media type is RJ45
  ARP type: ARPA, ARP Timeout 04:00:00,
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 input packets with dribble condition detected
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 1 interface resets
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
    0 output buffer failures, 0 output buffers swapped out
GigabitEthernet0/0/1 is up, line protocol is up (connected)
  Hardware is Lance, address is 00e0.b000.e202 (bia 00e0.b000.e202)
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255

```

```

Router#show interfaces
^
% Invalid input detected at '^' marker.

Router#show interface
GigabitEthernet0/0/0 is up, line protocol is up (connected)
  Hardware is Lance, address is 000a.f320.be01 (bia 000a.f320.be01)
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Full-duplex, 100Mb/s, media type is RJ45
  ARP type: ARPA, ARP Timeout 04:00:00,
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 input packets with dribble condition detected
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 1 interface resets
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
--More--

```

```
Router#show interface
GigabitEthernet0/0/0 is administratively down, line protocol is down (disabled)
  Hardware is ISR4331-3x1GE, address is 0060.2f9a.ed01 (bia 0060.2f9a.ed01)
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive not supported
  Full Duplex, 100Mbps, link type is auto, media type is Auto Select
  output flow-control is on, input flow-control is on
  ARP type: ARPA, ARP Timeout 04:00:00,
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/375/0 (size/max/drops); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts (0 IP multicasts)
    0 runs, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
    0 watchdog, 1017 multicast, 0 pause input
    0 input packets with dribble condition detected
```

```
Router#show interface
GigabitEthernet0/0/0 is up, line protocol is up (connected)
  Hardware is Lance, address is 0004.9ac8.9401 (bia 0004.9ac8.9401)
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Full-duplex, 100Mb/s, media type is RJ45
  ARP type: ARPA, ARP Timeout 04:00:00,
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runs, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 input packets with dribble condition detected
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 1 interface resets
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
```

Task 7

```
Router#ping 192.168.1.11
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.11, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
```

```
Router>ping 192.168.1.12
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.12, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

Router>
```

```
Router>ping 192.168.1.13
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.13, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
```

```
Router>ping 192.168.1.14
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.14, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

Router>
```

Task 8

Switch 1

```
Switch>show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       0001.9629.0a18   DYNAMIC   Fa0/24
1       000a.f320.be01   DYNAMIC   Fa0/2
1       00e0.b000.e201   DYNAMIC   Fa0/1
1       00e0.b000.e202   DYNAMIC   Fa0/3
Switch>
```

Switch 2

```
Switch>show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       0002.4abe.4818   DYNAMIC   Fa0/24
1       0004.9ac8.9401   DYNAMIC   Fa0/4
1       0060.2f9a.ed02   DYNAMIC   Fa0/3
Switch>
```

Task 9

Switch 1

```
Switch>clear mac address-table
      ^
% Invalid input detected at '^' marker.

Switch>show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       0001.9629.0a18   DYNAMIC   Fa0/24
1       000a.f320.be01   DYNAMIC   Fa0/2
1       00e0.b000.e201   DYNAMIC   Fa0/1
1       00e0.b000.e202   DYNAMIC   Fa0/3
Switch>
```

Switch 2

```

Switch>clear mac address-table
      ^
% Invalid input detected at '^' marker.

Switch>show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       0002.4abe.4818    DYNAMIC   Fa0/24
1       0004.9ac8.9401    DYNAMIC   Fa0/4
1       0060.2f9a.ed02    DYNAMIC   Fa0/3
Switch>

```

Task 10

```

Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

```

Task 11

```

Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
%Invalid interface type and number
Router(config)#interface Vlan 1
Router(config-if)#ip address 192.168.1.11 255.255.255.0
      ^

Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface Vlan 1
Router(config-if)#ip address 192.168.1.12 255.255.255.0
      ^

```

```

Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface Vlan 1
Router(config-if)#ip address 192.168.1.13 255.255.255.0

```

```

Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface Vlan 1
Router(config-if)#192.168.1.14 255.255.255.0

```

Task 12

```

Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

```

Task 15

Yes, some MAC addresses reappear almost immediately. Network devices constantly send automated background traffic.